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Effect of Minimum Wage on Changes in the Thai Labor Market[1,2]

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Abstract

This study evaluates the effect of the minimum wage on changes in the Thai labor market from 2002 to 2010, when the real minimum wage gradually decreased, and 2011 to 2013 when the real minimum wage substantially increased. These changes include labor force participation, employment, dis-employment, weekly working hours, real hourly wages, real hourly total labor income, and various other types of income. This study uses the individual-level panel data generated from the Matched-Outgoing Rotation Group (Matched-ORG) of the Thai Labor Force Survey. We observed the negative effect of minimum wage on employment, where the elasticity was in the range of - 0.0029 to -0.0474. We also observed the dis-employment for the foreign workers. We found that firms adjust working hours and various types of income to mitigate minimum wage shock. We conclude that the competitive equilibrium theory can reasonably explain the effect of minimum wage on employment as well as the overall changes in the Thai labor market from 2002 to 2013.

JEL Codes: E24, J21, J64, D59

Keyword: Minimum Wage, Labor Earnings, Labor Force Participation, Employment, competitive equilibrium

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I. Introduction

The effect of the minimum wage on employment is a long-debated issue. One class of explanation relies on the prediction of a standard competitive equilibrium, in which the increase in the minimum wage would cause a reduction in employment. There is a vast literature that provides consistent empirical evidence with this prediction (Neumark, Schweitzer, & Wascher, 2004; Neumark & Wascher, 1992, 1994, 2008). Another explanation argues that with the lack of competition among firms in acquiring labor input, firms can have monopsony power to set wages below the marginal product of labor. The increase in the minimum wage then brings wages back to its competitive optimum. Importantly, it does not result in the reduction of employment. There is an extensive literature providing evidence in support of this explanation (Card & Krueger, 1994; Dickens, Machin, & Manning, 1999; Katz & Krueger, 1992; Okudaira, Takizawa, & Yamanouchi, 2019).

For Thailand, the existing literature mostly found a small (and sometimes positive) effect of minimum wage on employment. These studies provide an explanation based on monopsony theory (Ariga, 2015; Lathapipat & Poggi, 2016; Strobl & Walsh, 2016). Their empirical estimation relies on cross-sectional or provincial-level panel data. This paper contributes to the existing studies by re-examining the empirical evidence of the effect of minimum wage on employment using the individual-level panel data, i.e., the Matched-Outgoing Rotation Group (Matched-ORG). This data set was constructed from the Thai Labor Force Survey (LFS). From the year 2002 onward, the survey has implemented the outgoing rotation group (ORG) sampling procedure. This procedure allows researchers to match individuals in the Outgoing Rotation Group (ORG) between the two consecutive years. This matching produces the individual panel data where an individual appears for two years. The panel data allow researchers to perform the fixed-effect estimation.

This paper also evaluates many other dimensions of labor market outcomes such as dis-employment, hourly wages, hourly total labor income, weekly working hours, monthly overtime income, yearly bonus income, and various types of monthly in-kind income. The period of study covers two distinct periods, i.e., the period where the minimum wage gradually decreased from 2002 to 2011 and the period where the minimum wage substantially increased from 2012 to 2013 (see Figure 1). The authors consider the effect during both periods together when interpreting the results.

Contrary to existing literature, this study concludes that the competitive equilibrium theory can reasonably explain the effect of the minimum wage on the overall changes in the

Thai labor market. We explain that the reduction in the real hourly minimum wage from 2002 to 2011 induces an increase in labor demand. This reduction then results in an increase in employment during this period. However, the firm labor demand had decreased from 2012 to 2013 since the minimum wage had substantially increased. This reduction leads to decreasing employment.

Meanwhile, we also observed the significant dis-employment for foreign workers, who can be considered as low-skilled. The other changes, such as labor force participation, real hourly total labor income, and various other types of income, support the above explanation. We also discuss our results by comparing them to existing literature. The remainder of the paper is organized as follows. Section II presents the detail of minimum wage policy in Thailand. Section III presents the detail of empirical data and section IV presents an empirical model. The empirical results are presented in section V. Section VI discusses the empirical results and section VII concludes the paper.

II. Minimum wage policy in Thailand

Thailand has implemented the minimum wage policy since 1973[3]. There is an exemption for some groups of workers, such as government employees, government enterprise employees, part-time employees, and agricultural workers.

The wage committee is an organization that determines the minimum wage rate. It consists of representatives from three parties, including the government, firms, and workers. The committee takes many socio-economic factors into account in order to determine the minimum wage rate. These include an indicator of the cost of living, workers' standard of living, labor productivity, prices of goods and services, cost of production, business capacity, socio-economic conditions, inflation, and Gross Domestic Product: GDP (Wage Committee, 1993). Figure 1 presents the average real minimum wage by region in Thailand from 1976 to 2017. The graph shows that the minimum wage had gradually declined from 1994 to 2011, then it dramatically increased from 2011 to 2013. Importantly, from 2011 to 2013, the Thai government increased the minimum wage in every province to 300 baht nationwide[4]. This

[3] During 1973 to 1997 the minimum wage was enforced by the virtue of Revolutionary Council Announcement No.103 (1973) and Ministry of Interior Announcement on Minimum Wage Rate No.1 - No.22 (1997). From 1998, the minimum wage was enforced by virtue of Labor protection act (1998)

[4] The government first increased the minimum wage up to 300 baht in seven provinces, including Phuket, Bangkok, Nakhon Pathom, Nonthaburi, Pathum Thani, Samut Prakan, and Samut Sakhon in April 2012. Then the 300 Baht rate went into effect nationwide in January 2013 (Ministry of Labour, 2012).

increase is about 62 percent on average[5]. In an empirical estimation, we estimated the effect of the minimum wage during these two periods separately. In the next section, we describe the empirical data in detail.

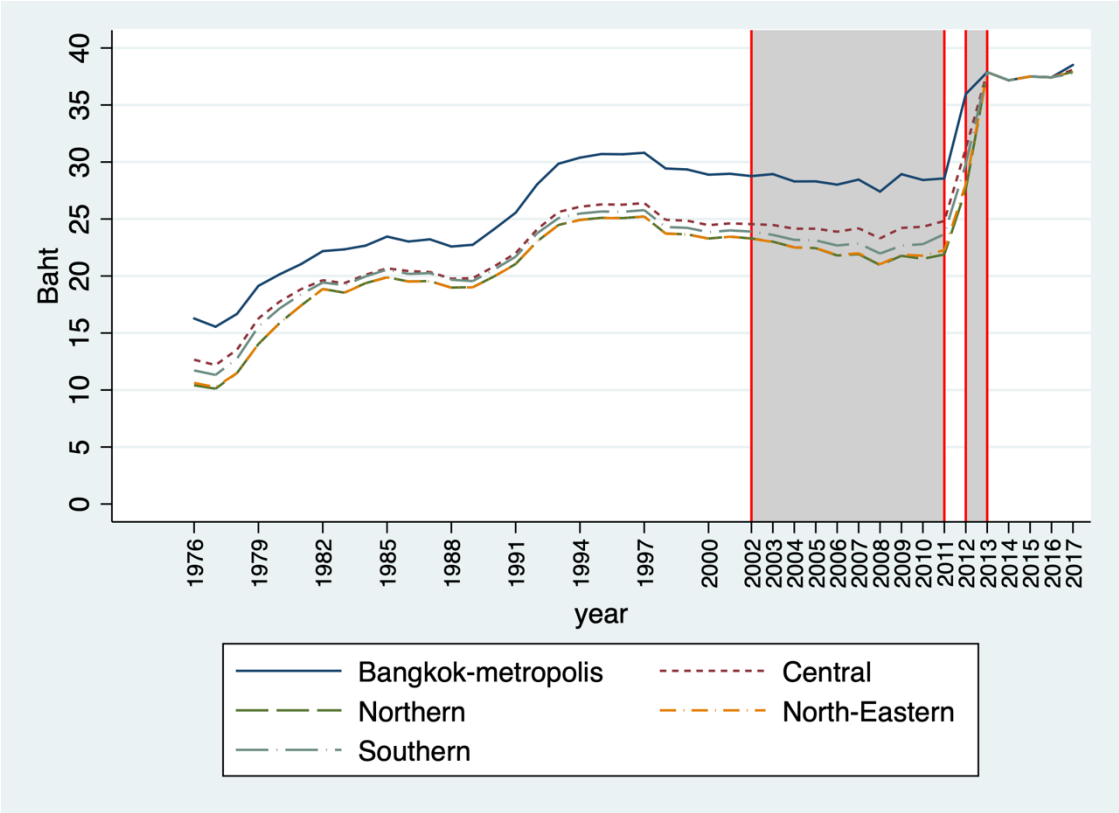

Figure 1 Average Real Hourly Minimum Wage by Region in Thailand from 1994 to 2017.

Notes: 1/ The minimum wage is adjusted to be real value using the regional consumer price index (CPI) with 2015 as a base year.

2/ The daily minimum wage rate is divided by 8 (hours) to get the hourly minimum wage rate.

III. Data

The data set used in this paper is the Matched-Outgoing Rotation Group (Matched-ORG). This panel data set was created from the Thailand Labor Force Survey (LFS). The National Statistical Office (NSO) initially designed the LFS data as cross-sectional data. However, from the year 2002 onward, the survey has implemented the outgoing rotation group

[5] The increase of real hourly minimum wage from 2012 to 2013 is 28.79 percent.

(ORG) sampling procedure. The ORG procedure first repeats the survey on the same household for two quarters. It will then pause for two consecutive quarters. After that, the survey will interview the same household again for two consecutive quarters. We call this procedure a 2-2-2 pattern.

The implementation of the ORG procedure provides an opportunity to match the same individual across time and create individual panel data. The empirical strategy to match the ORG sample is to link the individual using the household Identification Number (Household ID) and the individual's characteristics. We record individuals who live in the same household across consecutive years and have consistent characteristics to be the same person. These characteristics include gender, age, years of schooling, and marital status. We can create panel data for the years 2002-2003, 2004-2005, 2006-2007, 2008-2009, 2010-2011, and 2012-2013[6]. Each panel data set will consist of two years, denoted as year 1 and year 2. For instance, data set in the year 2002-2003 will mark 2002 as year 1 and 2003 as year 2.

The Matched-ORG data set provided information on the wage rate and working hours per week. We used this information to compute the hourly wage. To deal with outliers, we dropped individuals whose annual wage growth exceeded 1000 percent[7]. The data set provides various kinds of income, including overtime, bonus, other money, and in-kind income[8] such as food, clothing, other goods, and housing. We aggregate this information together with wages to create total labor income variables. Since the monetary value of in-kind variables were self-estimated by individuals, we account for the possibility of bias by creating another variable for comparison. We created a labor income variable, which contains only the directly observed wages, overtime income, and bonus income. Note that we convert all the nominal value into real value using the regional consumer price index (CPI) with 2015 as a base year.

The Matched-ORG data set also provides information on whether an individual had worked in the past seven days before the interview date. The interviewer also asked everyone who has not worked during the period whether they have still received a salary from their job(s). If the worker has not received a salary, the interviewer then asked whether they had got a position to get back to work. We record workers who answer yes to any of the above questions as being employed. Otherwise, they will be counted as unemployed.

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[6] The Outgoing Rotation Group sampling procedure would utilize the sample set for two years then it would change their sample set to the new one. So we cannot connect the sample set between 2003 to 2004, 2005 to 2006, 2007 to 2008, 2011 to 2012.

[7] We follow Neumark et al. (2004) to exclude individuals with wage growth exceeding 1000 percent as they potentially represent the outliers.

[8] The interviewer ask respondents to estimate the value of in-kind income in terms of cash. It is possible that this self-estimate would contain some errors. We assume that the sum of errors across the sample equal to zero.

For those individuals who were unemployed, the interviewer asked whether they have searched or applied for jobs within 30 days before the interview date. We mark individuals who answer yes, as well as those who were working on their careers as labor force participants.

The National Statistical Office (NSO) had estimated the sample weight to use with the LFS data. The procedure starts with calculating the inverse of the probability of the individual being in the sample. This inverse of probability would suggest a rough number of people which the individual represents. Then the weight is adjusted for non-response. Finally, the NSO adjusted weight by other information such as population by groups of age, gender, region, and administrative region (National Statistical Office, 2016). This procedure is consistent with the methodology used in the United States Current Population Survey: CPS data. We simply used this weight as a sample weight for Matched-ORG data.

Note that the Matched-ORG data also includes foreigners who live in Thailand. We identify foreigners from the question asking whether workers had registered for a working permit. However, this question is available form 2010 onward. We cannot identify foreign workers before this period.

IV. Empirical Model

This section describes the empirical models, which used to evaluate the effect of the minimum wage on changes in the labor market. These changes include labor force participation, employment, dis-employment, weekly working hours, real hourly wages, real hourly total labor income, real hourly labor income, monthly overtime income, yearly bonus income, monthly other money income, and monthly in-kind income such as food, clothing, other goods, and housing.

There are two concerns regarding simple linear regression analysis given the nature of empirical data. First, we should not ignore the possibility that there might be a time-invariant individual's unobserved characteristics, C_i, that are correlated with the minimum wage. For example, an individual with high ability may tend to move into or stay in an area with high wage growth. In this case, the minimum wage would positively correlate with the individual's unobserved productive characteristics, e.g., ability. This problem can be written as an unobserved effects model as follows (Wooldridge, 2010) :

$$Y_{i,s,t} = MW_{s,t}\beta + C_i + u_{i,s,t}, \qquad (1)$$

where $Y_{\mathrm{i,s,t}}$ is the outcome variable for an individual i who live in province s in period t, $\mathrm{MW_{s,t}}$ is the real minimum wage of province s in period t, $\mathrm{C_i}$ denotes an unobserved individual's productive characteristic of an individual i, which is not varied over time, and $\mathrm{u_{i,s,t}}$ denotes an idiosyncratic error term.

The second concern is that there might be time-variant unobserved effects correlated with the minimum wage. This concern comes from the fact that the wage committee determines the minimum wages by considering essential economic factors, which normally are time-variant. These factors include labor productivity, inflation, economic growth, gross domestic product (GDP), and other relevant socio-economic factors. We can extend the model in equation (1) as follows:

$$Y_{\mathrm{i,s,t}} = \mathrm{MW_{s,t}}\boldsymbol{\beta} + \mathrm{C_i} + \mathrm{V_{s,t}} + \mathrm{u_{i,s,t}}, \tag{2}$$

where $\mathrm{V_{s,t}}$ denote time-variant unobserved effects, which can vary through time t and province s. Here is the correlation between a minimum wage, $\mathrm{MW_{s,t}}$, and the unobserved $\mathrm{C_i}$ or $\mathrm{V_{s,t}}$ that can potentially generate an endogeneity problem.

Using the Matched-ORG data should overcome the first concern because the Matched-ORG is an individual panel data set. This panel structure allows us to apply the first-difference across two periods to equation (2) to eliminate the time-constant unobservable, $\mathrm{C_i}$, as follows:

$$\Delta Y_{\mathrm{i,s}} = \Delta\mathrm{MW_s}\boldsymbol{\beta} + \Delta\mathrm{V_s} + \Delta\mathrm{u}_{i,s}, \tag{3}$$

where $\Delta\mathrm{Y_{i,s}} = \mathrm{Y_{i,s,t}} - \mathrm{Y_{i,s,t-1}}$, $\Delta\mathrm{MW_s} = \mathrm{MW_{s,t}} - \mathrm{MW_{s,t-1}}$, $\Delta\mathrm{V_s} = \mathrm{V_{s,t}} - \mathrm{V_{s,t-1}}$ and $\Delta\mathrm{u_{i,s}} = \mathrm{u_{i,s,t}} - \mathrm{u_{i,s,t-1}}$. An endogeneity problem still can arise, however, when there is a correlation between the change in the minimum wage, $\Delta\mathrm{MW_s}$, and changes in time-variant unobserved effect $\Delta\mathrm{V_s}$. We account for this problem by focusing on the data from 2012 to 2013. During this period, the Thai government increased the minimum wage to reach the same rate at 300 Baht in every province. This statutory rate was not conditioned on provincial socioeconomics as has been done before. We assume that the changes in minimum wage during this period, $\Delta\mathrm{MW}_s$ is uncorrelated with the change in time-variant unobserved effect, $\Delta\mathrm{V_s}$. This assumption is to solve the second concern. Then we can obtain a consistent estimation of $\boldsymbol{\beta}$.

Equation (4) presents our first empirical model, which follows Neumark et al., (2004). The dependent variable $\frac{Y_{i,s,t=2}-Y_{i,s,t=1}}{Y_{i,s,t=1}}$ denotes the percentage change of an outcome variable for an individual i from year 1 to year 2, including weekly working hours, real hourly total labor income, real hourly labor income, real hourly wage, real monthly other money income, real yearly bonus income, real monthly in-kind income such as food, clothing,

$$\frac{Y_{i,s,t=2} - Y_{i,s,t=1}}{Y_{i,s,t=1}} = \alpha + \sum_{j} \beta_j \frac{MW_{s,t=2} - MW_{s,t=1}}{MW_{s,t=1}} . R_j(W_{i,s,t=1}, MW_{s,t=1})_j + \sum_{j} \beta_j^L \frac{MW_{s,t=1} - MW_{s,t=0}}{MW_{s,t=0}} . R_j(W_{i,s,t=1}, MW_{s,t=1})_j + \sum_{j} \gamma_j R_j(W_{i,s,t=1}, MW_{s,t=1})_j + \sum_{j} \phi_j R_j(W_{i,s,t=1}, MW_{s,t=1})_j . \frac{W_{i,s,t=1}}{MW_{s,t=1}} + X_{i,s,t=1}\delta + D_{i,t=1}^{M}\lambda + D_{i,t=1}^{S} * D_{i,t=1}^{Y}\pi + \varepsilon_{i,s,t} \quad (4)$$

Table 1 Wage position of an individual described by the ratio between an individual's real hourly wage and real hourly minimum wage

Index of wage position; j	Range of ratio of real hourly wage compared to real hourly minimum wage
1	W/MW ≤ 0.9
2	0.9 < W/MW ≤ 1.1
3	1.1 < W/MW ≤ 1.2
4	1.2 < W/MW ≤ 1.3
5	1.3 < W/MW ≤ 1.5
6	1.5 < W/MW ≤ 2
7	2 < W/MW ≤ 3
8	3 < W/MW ≤ 4
9	4 < W/MW ≤ 5
10	5 < W/MW ≤ 6
11	6 < W/MW ≤ 8

other goods, and housing. The variable $\frac{MW_{s,t=2}-MW_{s,t=1}}{MW_{s,t=1}}$ denotes the percentage change in real hourly minimum wage from year 1 to year 2, respectively; the variable, $R_j(W_{i,s,t=1}, MW_{s,t=1})_j$ denotes dummy variable indicates an individual's wage position

compared to the minimum wage (see Table 1 for details). Importantly, β_j, is the coefficient of the interaction term that measures the effect of the minimum wage according to individual wage position. $\frac{MW_{s,t=1}-MW_{s,t=0}}{MW_{s,t=0}}$ denotes the percentage change in real hourly minimum wage from year 0 (i.e., the year before year 1) to year 1; $\frac{W_{i,s,t=1}}{MW_{s,t=1}}$ denotes the ratio between real hourly wage and real hourly minimum wage; $X_{i,s,t=1}$ denotes a set of controlled variables on individual characteristics, including dummy variable indicating female, years of schooling, years of potential working experience, and its square term; $D_i^S * D_i^Y$ denotes the interaction term between the dummy variable indicating province and year, respectively; D_i^M denotes the dummy variable indicating the month of the survey; and $\varepsilon_{i,s,t}$ denotes an idiosyncratic error term. We also apply the model in equation (4) to an individual's dis-employment. By conditioning on individuals who are currently employed in year 1, we observed whether the individual turns to be un-employed in year 2. This outcome variable for dis-employment denoted as a dummy variable $D_{i,s,t}^{DE}$, which has a value of 1 if an individual turns to be un-employed in year 2, and 0 otherwise.

Our second empirical model is to estimate the effect of the minimum wage on labor force participation and employment. Equation (5) below presents the model as follows:

$$E_{i,s,t} = \eta + MW_{s,t}\theta + X_{i,s,t}\delta + D_i^S\rho + D_i^Y\pi + \epsilon_{i,s,t}, \quad (5)$$

where $E_{i,s,t}$ denotes a dummy variable that indicates an individual's outcome, i.e., labor force participation, or employment status. The labor force participation (employment or seeking for jobs) variable has the value of one if the individual i, who lives in province s, is participating in the labor force (employed) in year t and value of zero otherwise; $MW_{s,t}$ denotes real hourly minimum wage in province s and year t; $X_{i,s,t}$ denotes a set of control variables on individual characteristics, including the gender of female, years of schooling, years of working experience, and its square term; D_i^Y and D_i^S denote dummy variables on year and province, respectively; and $\epsilon_{i,s,t}$ indicates an idiosyncratic error term. We estimate equation (5) using the probit population-average model with the Generalized Estimating Equations (GEE) approach (Wooldridge, 2010; Zeger, Liang, & Albert, 1988) in order to deal with a potential endogeneity problem.

From Figure 1, the real minimum wage had gradually decreased from 2002 to 2011. However, the real minimum wage substantially increased from 2012 to 2013. We account for these different trends by estimating the empirical models in equations (4) and (5) for these two periods separately.

V. Empirical Results

In this section, we present the effect of real hourly minimum wage on various labor market outcomes. To interpret the effect on various kinds of incomes and dis-employment, we focus on the coefficient, β_j, in equation (4). When we interpret the effect of real hourly minimum wage on labor-force participation and employment, we focus on the coefficient θ in equation (5).

For illustrative purpose, we summarized some essential estimation in a graphical illustration shown in Figure 3, Figure 4, and Figure 5. The graph showing the coefficients β_j and their 90 percent confidence interval for real hourly wages, real hourly total labor income, real monthly overtime income, weekly working hours, and dis-employment from 2002 to 2011 and 2012 to 2013, respectively.

A. Period of 2002 to 2011

Note from Figure 1 that the real value of hourly minimum wage has declined during this period. The results in this section describe the effect of this decline in real hourly minimum wage on the various types of labor market outcomes. The first outcome of interest is the change in real hourly wages. Figure 3, panel A (or see Table 2, column 1) shows that the coefficients β_j have a positive sign throughout the wage distribution. The coefficients are also statistically significant across the wage distribution. These results suggest that real hourly wages and real hourly minimum wages changed in the same direction. Since the real hourly minimum wages had decreased from 2002 to 2011, we interpret that the real hourly wages also declined during this period.

We found the mixed effect on real yearly bonus income, real monthly other money income, and real monthly in-kind income. Table 2, column 6 to 10, shows the coefficients with a mixed effect between positive and negative signs. However, the effects are weaker compared to the effects on wages; that is, the coefficients are not statistically significant.

Despite the reduction in wages and various types of income, we found an increase in weekly working hours and real monthly overtime income. The results showing in Figure 3, panel C, and D (or see Table 2, columns 3 and 4) indicate that the coefficients have a negative sign and are statistically significant. The negative coefficients suggest an increase in these two outcomes, in contrast with the reduction of the real hourly minimum wage during this period.

We interpret that the firms realize the decline in real hourly minimum wages even though its nominal value had not decreased during this period. The high inflation rate had blinded its reduction in real value (see Figure: 2 for the comparison of the nominal and real value of the minimum wage from 1975 to 2017). So, firms acquire more units of labor, i.e., working hours, in response to a decrease in labor input price.

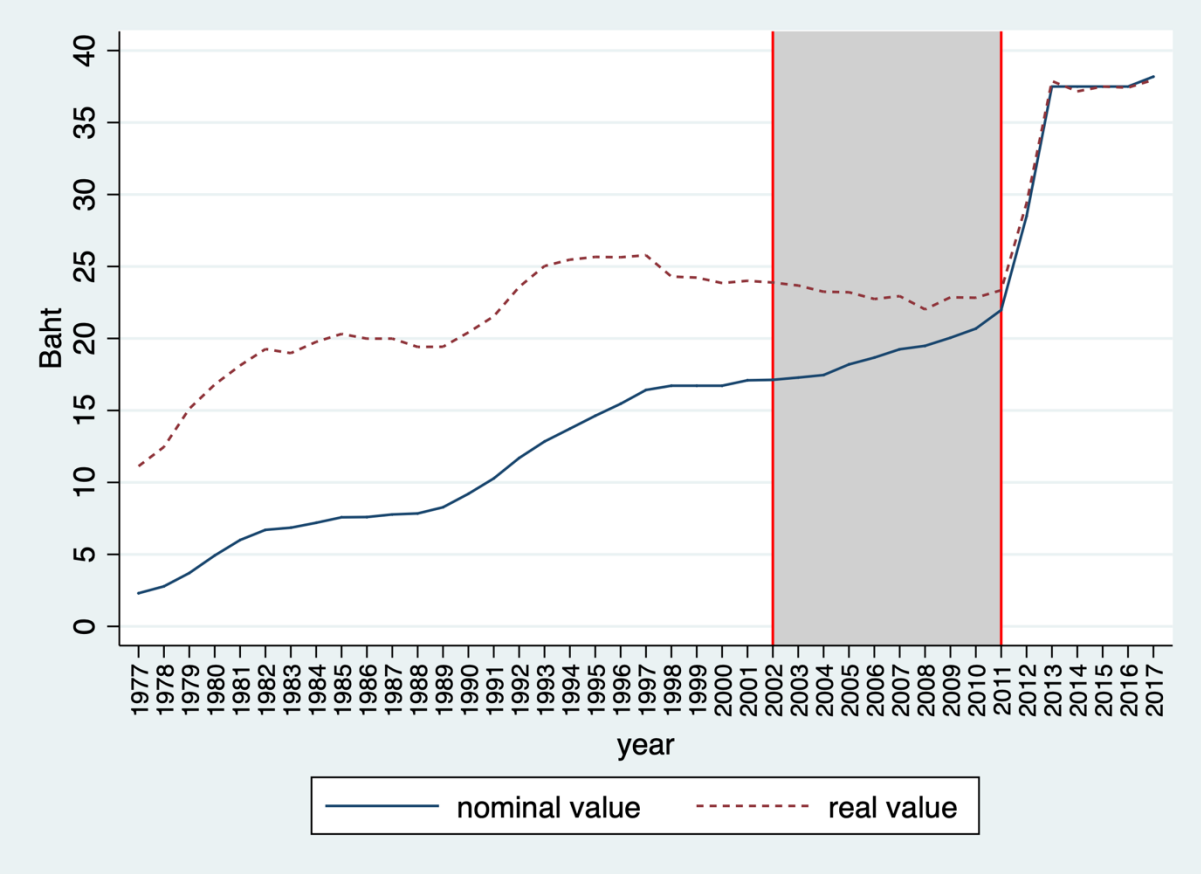

Figure: 2 Nominal and real hourly minimum wage

Note: The authors converted the nominal to real value by using the consumer price index with 2015 as a base year.

This mechanism leads to an increase in real monthly overtime income. The increase in real monthly overtime income is so much that it can maintain the real hourly total labor income. We found negative coefficients on these two variables for the workers who earn around and above the minimum wage, though not statistically significant. This result, nevertheless,

suggests that real hourly total labor income did not decline in response to the reduction of the real hourly minimum wage during this period[9].

The first column in Table 4 presents the effect of minimum wages on labor force participation from 2002 to 2011. The coefficient is negative, however, not statistically significant. The negative sign suggests that labor force participation did not decrease, even though the real hourly minimum wages had decreased during this period. This result is counter-intuitive at first. We interpret that the labor force participation represents the worker labor supply, a function of real hourly total labor income (not just the wage) as the price of labor input. So, we do not observe the decrease in labor supply since the price of labor had not decreased as described above.

Surprisingly, we also found an increase in employment. The first column in Table 5[10] shows that the coefficient is negative and statistically significant. This result suggests that employment had increased, in contrast with the reduction in real hourly minimum wage during this period.

Table 7 presents the effect of minimum wages on dis-employment. This effect determines whether a worker will turn to be unemployed due to the change in the minimum wage. The first column presents the effects from 2002 to 2011. The coefficient is positive, which implies that the dis-employment had decreased. The effects are statistically significant for workers who earn at the minimum wage rate and for many other wage positions. This finding is consistent with results on employment, where we found an increase in employment during this period.

B. Period of 2012 to 2013

Figure 4, panel A (or see Table 3, column 1) shows that the coefficients for wages are positive throughout the wage distribution. This positive result is consistent with the prior period, where we saw wages and minimum wages move in the same direction. The effect was significant for workers who earn at or around the minimum wage, as well as those with wages higher up to 1.5 times the minimum. This result is in contrast with the prior period, where we found significant wage changes across the distribution.

[9] The increasing in overtime can compensate for the reduction in wage rate since the hourly overtime rate is 1.5 times higher than the standard 8-hours rate (Labor protection act, 1998).
[10] see also Table 6 for the converted elasticity

In contrast to an increase in wages, we found a reduction in various types of incomes, including other money income, in-kind income such as food, and other goods. Table 3, column 7, 8, and 9 show the coefficients with a negative sign, but not statistically significant. We interpret that firms tend to reduce these types of payments to minimize their cost, in response to a jump in the minimum wages. However, we found a positive coefficient for real yearly bonuses and mixed signs for real monthly housing but not statistically significant. We do not have an explanation for this phenomenon, however.

Figure 4, panel C and D (or see Table 3; columns 4 and 5) indicate that while the weekly working hours increased, the real monthly overtimes income decreased. These results are counter-intuitive at first. We interpret that firms may comply with paying the minimum wage rates for standard 8-hour work. However, it may negotiate with workers to work an extra-hour without receiving an overtime income. Even though doing so is illegal. The firms may have done so to mitigate the shock on labor costs. It turns out that the firm can maintain the labor price. The coefficients for real hourly total labor income in panel B, Figure 4 (or see Table 3 column 3) are positive, though not statistically significant.

Table 4 presents the effect of changes in real hourly minimum wages on labor force participation. The second and third columns present results for Thai and overall workers from 2012 to 2013[11]. The results suggest that the labor force participation for Thai workers and the overall workers had increased. The increase in labor force participation for Thai and overall workers is consistent with the increase in the real hourly minimum wage and the real hourly total labor income. We interpret that an individual's labor supply had increased in response to the increase in the minimum wage.

Despite an increase in labor force participation, we found a sign of reduction in employment for both Thai and foreign workers. Table 5, from the second to the fourth column, show that the coefficients are negative, but not statistically significant. This result is in contrast with the prior period, where we found the labor force participation and employment move in the same direction. We interpret that during this period, the dramatic increase in the real minimum wage had reduced the firm's labor demand. So, *equilibrium* employment occurs at a lower level.

Further evidence on dis-employment shows that the foreign workers had become unemployed during this period. The coefficients are positive and statistically significant (see the third column of Table 7). Surprisingly, there was no dis-employment for Thai workers,

[11] In this period, we can identify whether workers are Thais or foreigners. But the number of observation is not enough for estimating the results.

where the coefficients in the second column are negative. We interpret that when the minimum wage dramatically increased, firms tended to hire Thai workers rather than foreigners, given the same minimum wage rate since Thai workers are endowed with higher skills, e.g., better in Thai language and literacy skills. Lastly, we did not find a dis-employment effect on the overall labor market. The last column of Table 7 shows the coefficient with a negative sign, but not statistically significant.

VI. Discussion

In the previous section, we have seen the effect of the minimum wage on various types of labor market outcomes. Our data allow us to observed changes during the two main periods, the period where minimum wage gradually decreased from 2002 to 2011 and the period where minimum wage substantially increased from 2012 to 2013. We have two main issues to discuss in this section.

The first issue is that both firms and workers have incentives to maximize their benefits, given the minimum wages. During this period, the reduction in real minimum wages is associated with the reduction in real wages across the wage distribution. This phenomenon does not occur during the period of the real minimum wage increase, since doing so does not satisfy the firms' profit maximization. The firm also adjusts various types of in-kind payments, which are not regulated by law, in order to minimize their labor cost. We observed that these types of compensations tend to reduce, regardless of the trend of the minimum wage.

Firms also responded to the reduction in the real minimum wage from 2002 to 2011 by purchasing more labor. The increase in employment, working hours, and overtime income confirm this interpretation. We argue that the increased firm's labor demand can be supported by the increase in labor supply since the real hourly total labor income does not fall. We view the real hourly total labor income as the *true* wage, which workers considering when making labor supply decisions. This interpretation is consistent with the results of labor force participation and dis-employment during this period.

The different adjustment occurs when the real minimum wage substantially increased from 2012 to 2013. We observed a sign of a reduction in employment both for Thai and foreign workers. We interpret this because the firm’s labor demand had decreased during this period. However, if we consider things from the worker perspective, the increase in the real minimum wage should increase an individual’s willingness to work, i.e., labor supply. Our evidence had

supported this argument. We observed an increase in labor force participation both for Thai and overall workers during this period.

Despite the negative effect of the minimum wage on employment, our study points out to its small effect in size. Table 6 present the minimum wage elasticity of employment. The results show that the effects were small and insignificant, especially during the big jump in the minimum wage. The elasticities were in the range of -0.0029 to -0.0474 during both periods. This result is consistent with recent studies that found the small effect of the minimum wage on employment in Thailand (del Carpio, Messina, & Sanz-de-Galdeano, 2014; Lathapipat & Poggi, 2016). In addition to the local studies, the literature also found a small effect on employment. This phenomenon occurs alongside various kinds of firm internal adjustment, including increasing product price, cutting non-labor cost, improving productivity and substituting labor with capital (Bodnár et al., 2018; Caliendo, Fedorets, Preuss, Schröder, & Wittbrodt, 2018; Harasztosi & Lindner, 2019; Hirsch, Kaufman, & Zelenska, 2015).

Our study also suggests that layoffs might not be the firm's most favorite choice to deal with the minimum wage increase. We did not observe the significant effect of dis-employment on the overall worker, especially during the big jump in the minimum wage. This result is consistent with recent studies that observed an adjustment in employment flow, not stock (Bodnár et al., 2018; Caliendo et al., 2018; Dube, William Lester, & Reich, 2016; Meer & West, 2016). However, we observed the dis-employment for foreign workers, but not for Thais. We interpret that foreign workers might be more low-skilled and therefore more dispensable. This result is consistent with recent literature that found an adverse effect of the minimum wage on the low-skilled group (Clemens & Wither, 2019; Lordan & Neumark, 2018).

The second issue is to argues that the competitive equilibrium theory can reasonably explain the effect of minimum wage on the overall Thai labor market. This conclusion is different from the existing literature, which provides an explanation based on monopsony theory (Ariga, 2015; Lathapipat & Poggi, 2016; Strobl & Walsh, 2016). However, our pieces of evidence do not rule out the possibility in which the monopsony occurs in the sub-sector of the economy. There are recent studies in the United States and Japan estimating the heterogeneous effect of the minimum wage among the market and states (Okudaira et al., 2019; Wang, Phillips, & Su, 2019). This issue is still open for future research in the Thai labor market.

VII. Conclusion

This study evaluates the effect of the minimum wage on changes in the Thai labor market from 2002 to 2013. These changes include labor force participation, employment, dis-employment, weekly working hours, real hourly wages, real hourly total labor income, and various other types of income. This study contributes to the existing literature by utilizing the newly created Matched-Outgoing Rotation Group (Matched-ORG), which is the individual level panel data. The study covers the period where the real minimum wage gradually decreased from 2002 to 2011, and the period where the real minimum wage substantially increased from 2012 to 2013.

We found that while the minimum wage decreased, employment had increased. We argue that the firms' labor demand had increased during this period. However, we found a reduction in employment for Thai and foreign workers during the big jump in the minimum wage. The size of the effects is small and indifferent from zero. We also observed the significant dis-employment for foreign workers, which indicates the adverse effect of the minimum wage on the low-skilled group. Moreover, we also observed the increase in labor force participation, which indicates increases in workers' labor supply. We conclude that the competitive equilibrium theory can reasonably explain the effect of the minimum wage on employment as well as the overall changes in the Thai labor market from 2002 to 2013.

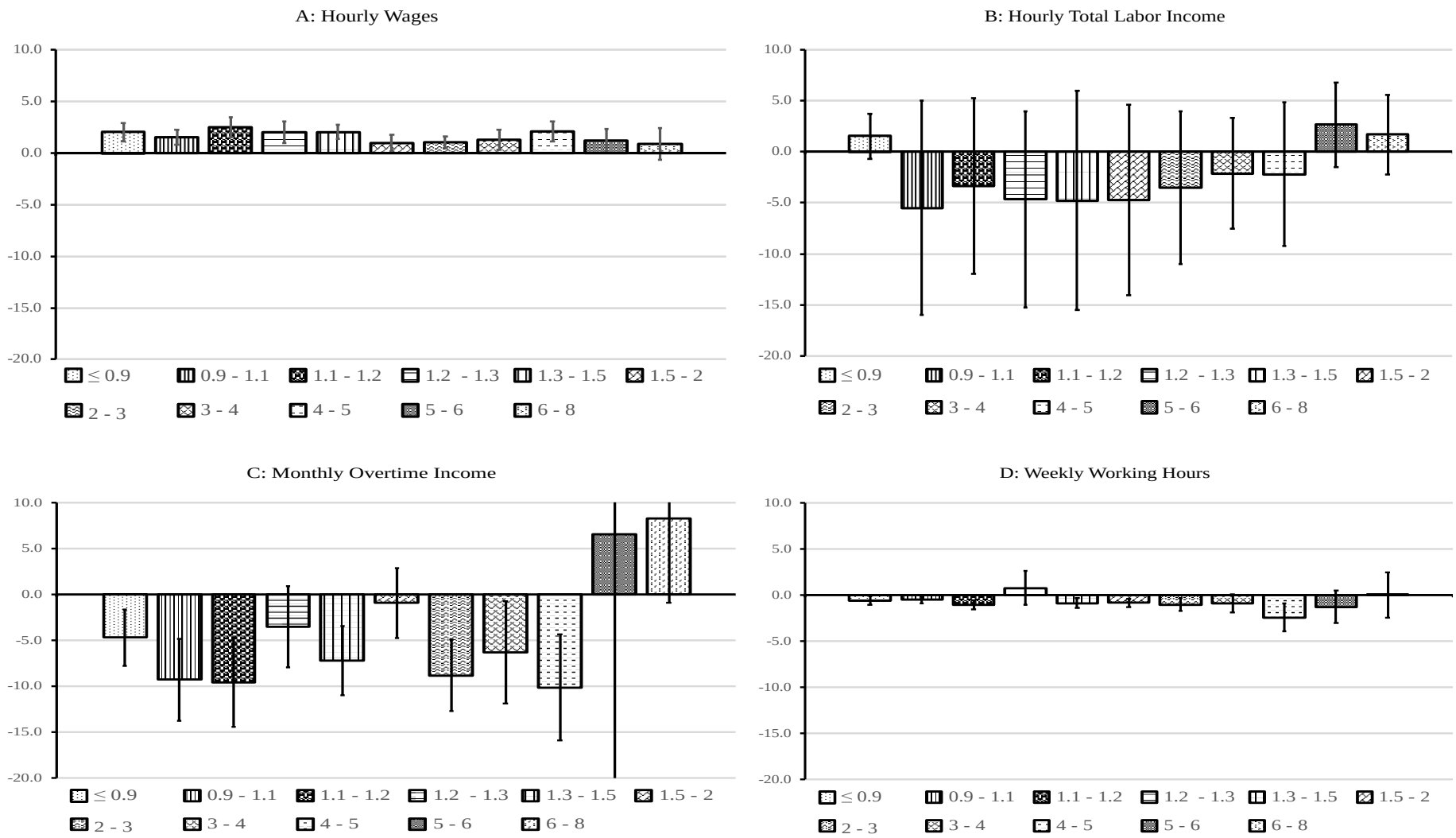

Figure 3 Coefficient, β_j, and 90 percent confidence interval for real hourly wages, real hourly total labor income, real monthly overtime income, and weekly working hours from 2002 to 2011

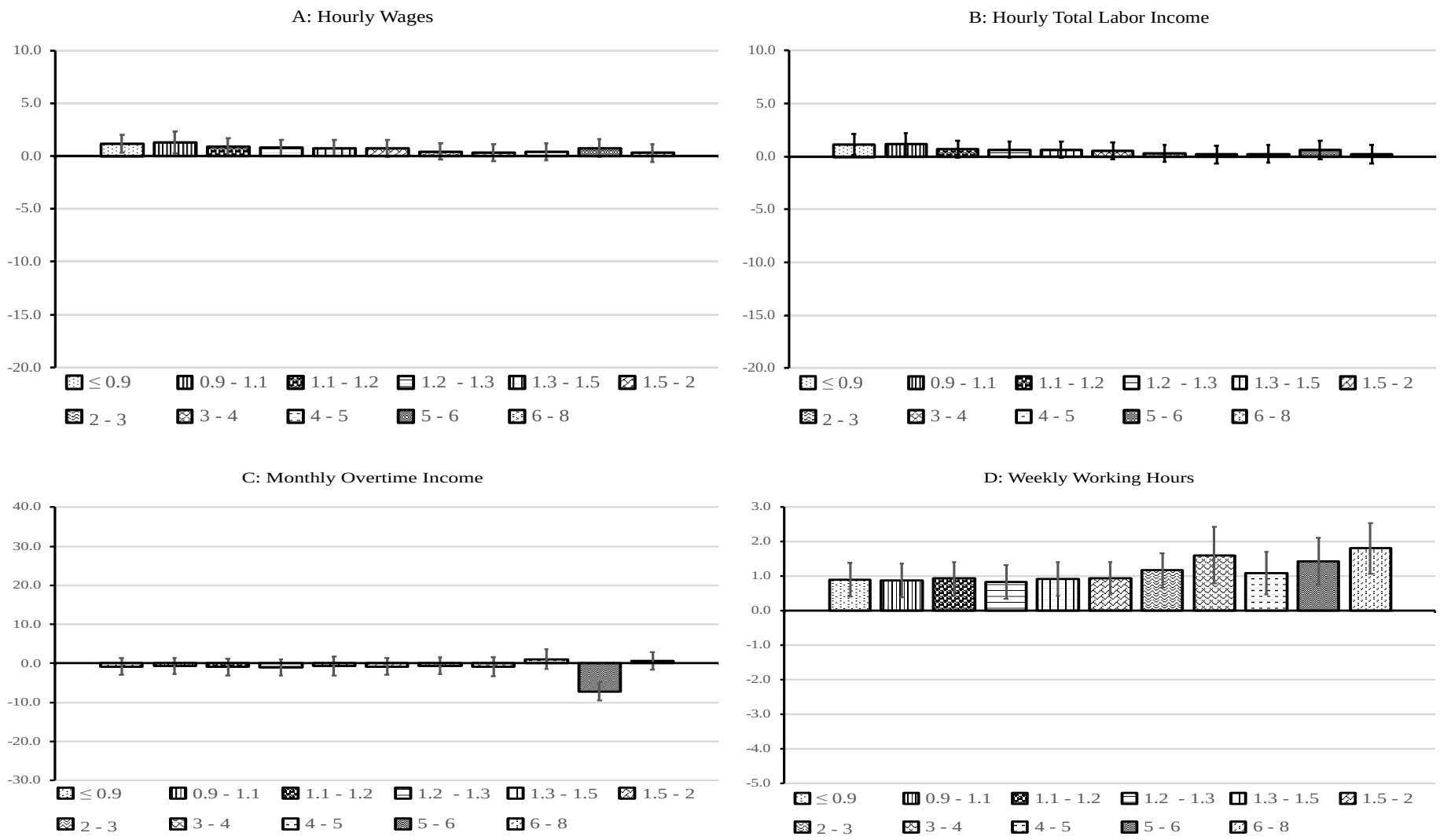

Figure 4 Coefficient, β_j, and 90 percent confidence interval for real hourly wages, real hourly total labor income, real monthly overtime income, and weekly working hours from 2012 to 2013

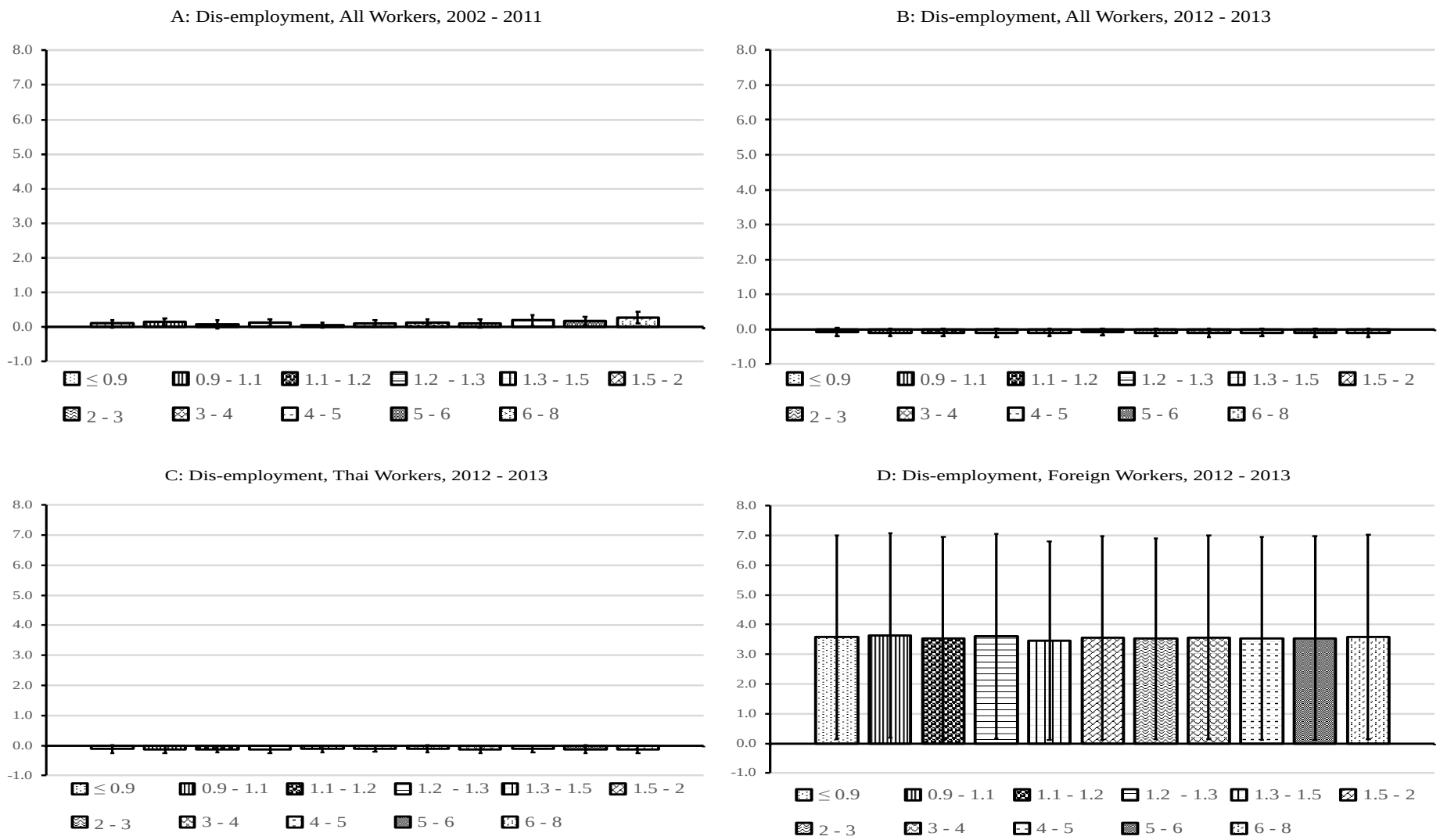

Figure 5 Coefficient, β_j, and 90 percent confidence interval for dis-employment

Table 2 Effects of the minimum wages on working hours and wages, total labor income, overtime income, bonus income, other money income, and in-kind income such as food, clothing, other goods, and housing from 2002 to 2011

Percent change in the minimum wage x Dummy variables for	Effect of the Minimum Wages, β_j, on										
	Hourly Wages	Hourly Labor Income	Hourly Total Labor Income	Working Hours	Overtime Income	Bonus Income	Other Money Income	In-kind Income			
								Food	Clothing	Other Goods	Housing
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
W/MW ≤ 0.9	2.039***	1.604	1.532	-0.552*	-4.684**	8.317	3.294	3.349	4.579	-0.120	-0.391
	(0.545)	(1.373)	(1.359)	(0.303)	(1.868)	(10.86)	(2.264)	(4.620)	(4.297)	(2.141)	(2.008)
0.9 < W/MW ≤ 1.1	1.528***	-5.585	-5.495	-0.486*	-9.263***	0.783	4.848**	5.957	7.558	-1.268	0.241
	(0.445)	(6.422)	(6.373)	(0.254)	(2.708)	(9.061)	(2.014)	(5.834)	(9.805)	(1.712)	(2.458)
1.1 < W/MW ≤ 1.2	2.486***	-3.517	-3.381	-1.080***	-9.544***	13.74	0.175	4.589	7.507	2.353	2.081
	(0.574)	(5.335)	(5.239)	(0.299)	(2.981)	(9.479)	(2.591)	(3.966)	(8.040)	(2.072)	(3.398)
1.2 < W/MW ≤ 1.3	1.996***	-4.781	-4.655	0.780	-3.508	10.05	5.743	0.697	16.25	1.139	3.355
	(0.649)	(6.523)	(6.480)	(1.111)	(2.675)	(10.06)	(3.662)	(3.189)	(15.46)	(2.216)	(2.442)
1.3 < W/MW ≤ 1.5	2.047***	-4.910	-4.778	-0.855**	-7.209***	6.843	1.738	3.021	13.95	0.0546	-5.615**
	(0.402)	(6.569)	(6.531)	(0.339)	(2.274)	(9.943)	(2.342)	(3.398)	(9.965)	(2.353)	(2.742)
1.5 < W/MW ≤ 2	0.966**	-5.049	-4.723	-0.822***	-0.918	23.87	1.706	-0.127	-5.409	-1.643	0.631
	(0.479)	(5.714)	(5.657)	(0.273)	(2.334)	(22.92)	(2.284)	(3.121)	(5.156)	(1.935)	(2.102)

Table 2 (continued)

Percent change in the minimum wage x Dummy variables for	Effect of the Minimum Wages, β_j, on										
	Hourly Wages	Hourly Labor Income	Hourly Total Labor Income	Working Hours	Overtime Income	Bonus Income	Other Money Income	In-kind Income			
								Food	Clothing	Other Goods	Housing
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
2 < W/MW ≤ 3	1.038***	-3.741	-3.522	-1.019**	-8.802***	1.537	-0.290	1.988	-3.823	0.261	3.904
	(0.363)	(4.607)	(4.559)	(0.409)	(2.346)	(10.14)	(2.554)	(3.307)	(4.532)	(1.602)	(2.781)
3 < W/MW ≤ 4	1.313**	-2.175	-2.112	-0.857	-6.311*	1.962	-0.493	-4.035	-7.334	-2.328	3.669
	(0.588)	(3.298)	(3.313)	(0.600)	(3.382)	(9.085)	(2.586)	(4.406)	(7.034)	(2.090)	(3.255)
4 < W/MW ≤ 5	2.099***	-2.361	-2.207	-2.413***	-10.12***	8.591	5.021*	2.429	29.29*	1.488	-0.811
	(0.584)	(4.359)	(4.294)	(0.913)	(3.502)	(11.19)	(2.567)	(3.372)	(15.70)	(1.813)	(4.785)
5 < W/MW ≤ 6	1.183*	2.209	2.656	-1.261	6.559	12.31	3.819	5.438	-4.801	-7.338**	-6.930
	(0.691)	(2.489)	(2.523)	(1.073)	(17.28)	(10.18)	(3.833)	(6.178)	(10.11)	(3.495)	(4.678)
6 < W/MW ≤ 8	0.893	1.554	1.698	0.0482	8.313	49.07	10.01	-1.546	8.630	5.848	-4.720
	(0.920)	(2.366)	(2.367)	(1.494)	(5.573)	(34.38)	(6.613)	(5.060)	(21.73)	(8.017)	(3.626)
Observations	77,964	77,997	77,905	79,910	9,141	5,733	10,481	8,829	2,946	3,756	3,489
Adj. R-Square	0.180	0.001	0.000	0.102	0.142	0.284	0.069	0.036	0.045	0.149	0.094

Notes: [1/] A superscript *, **, and *** denote p < 0.1, 0.05, and 0.01 respectively.

Table 3 Effects of the minimum wages on working hours and wages, labor income, overtime income, bonus income, other money income, and inkind income such as food, other goods, and housing from 2012 to 2013

Percent change in the minimum wage x Dummy variables for	Effect of the Minimum Wages, β_j, on									
	Hourly Wages	Hourly Labor Income	Hourly Total Labor Income	Working Hours	Overtime Income	Bonus Income	Other Money Income	In-kind Income		
								Food	Other Goods	Housing
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
W/MW ≤ 0.9	1.152**	0.889*	1.125*	0.908***	-0.775	11.73	-3.371	-2.053	-2.869	2.239
	(0.519)	(0.515)	(0.588)	(0.298)	(1.285)	(18.01)	(5.760)	(2.456)	(6.415)	(3.523)
0.9 < W/MW ≤ 1.1	1.287**	1.007	1.145*	0.880***	-0.645	11.74	-7.087	-1.205	-5.968	3.714
	(0.645)	(0.643)	(0.661)	(0.300)	(1.279)	(14.45)	(6.233)	(2.604)	(7.383)	(3.485)
1.1 < W/MW ≤ 1.2	0.892*	0.591	0.710	0.935***	-0.927	1.586	-5.824	-1.547	-7.460	-0.119
	(0.468)	(0.461)	(0.486)	(0.287)	(1.357)	(15.77)	(6.153)	(2.533)	(7.740)	(3.431)
1.2 < W/MW ≤ 1.3	0.783*	0.536	0.636	0.839***	-1.024	14.15	-10.18	-2.873	-0.998	-0.454
	(0.450)	(0.446)	(0.466)	(0.293)	(1.292)	(15.66)	(9.686)	(4.322)	(8.849)	(3.015)
1.3 < W/MW ≤ 1.5	0.769*	0.517	0.645	0.917***	-0.648	10.89	-6.743	-2.670	-5.028	-0.441
	(0.461)	(0.455)	(0.479)	(0.295)	(1.455)	(14.89)	(6.414)	(2.453)	(6.666)	(3.690)
1.5 < W/MW ≤ 2	0.732	0.454	0.542	0.941***	-0.764	7.082	-4.975	-1.992	-3.758	2.903
	(0.466)	(0.459)	(0.483)	(0.286)	(1.336)	(15.89)	(5.761)	(2.556)	(6.699)	(3.525)

Table 3 (continue)

Percent change in the minimum wage x Dummy variables for	Effect of the Minimum Wages, β_j, on									
	Hourly Wages	Hourly Labor Income	Hourly Total Labor Income	Working Hours	Overtime Income	Bonus Income	Other Money Income	In-kind Income		
								Food	Other Goods	Housing
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
2 < W/MW ≤ 3	0.440	0.186	0.314	1.171***	-0.592	9.420	-11.40	-1.264	-4.908	2.726
	(0.462)	(0.455)	(0.479)	(0.307)	(1.318)	(14.32)	(9.278)	(2.381)	(6.869)	(3.301)
3 < W/MW ≤ 4	0.351	0.0641	0.183	1.614***	-0.799	3.534	-5.863	-2.599	-3.323	2.138
	(0.497)	(0.492)	(0.519)	(0.502)	(1.501)	(13.41)	(6.611)	(2.514)	(6.592)	(3.646)
4 < W/MW ≤ 5	0.389	0.162	0.223	1.097***	1.123	4.582	-2.446	-1.450	-6.547	3.974
	(0.486)	(0.479)	(0.505)	(0.376)	(1.529)	(15.53)	(6.886)	(2.459)	(9.723)	(3.607)
5 < W/MW ≤ 6	0.755	0.482	0.600	1.439***	-7.176***	5.143	-10.27	-6.656	-17.89**	486.5
	(0.513)	(0.509)	(0.533)	(0.417)	(1.440)	(12.14)	(10.19)	(4.294)	(8.877)	(311.3)
6 < W/MW ≤ 8	0.303	0.0632	0.194	1.809***	0.696	2.511	-7.602	-18.47	-9.160	na
	(0.522)	(0.516)	(0.544)	(0.448)	(1.368)	(13.58)	(8.465)	(15.99)	(9.495)	
Observations	7,329	7,324	7,310	7,466	939	148	1,013	666	292	210
Adj. R-Square	0.274	0.257	0.229	0.130	0.113	0.230	-0.126	0.414	0.188	0.279

Notes: 1/ The superscript *, **, and *** denote p < 0.1, 0.05, and 0.01 respectively.

2/ na indicates that the estimation result is not available due to the insufficient number of observations in that wage position.

3/ We omitted the estimation result on clothing due to the insufficient number of observations.

Table 4 Effect of the minimum wage on labor force participation

Explanatory Variables	Dependent Variable: Labor Force Participation		
	2002-2011	2012-2013	
		Thai Workers	Overall
Minimum Wage	-0.0789	0.0403*	0.0370*
	(0.0706)	(0.0207)	(0.0201)
Female	0.0632	0.0954	0.0897
	(0.0577)	(0.168)	(0.150)
Years of Schooling	0.0439***	0.0247	0.0195
	(0.0132)	(0.0250)	(0.0222)
Potential Working	0.0313***	0.0649***	0.0628***
Experience	(0.00861)	(0.0151)	(0.0139)
Potential Working Exp.^2	-0.000493***	-0.000876***	-0.000856***
	(0.000118)	(0.000232)	(0.000213)
Foreigner			0.832***
			(0.232)
Observations	135,413	16,455	19,395
Number of Panel ID	34,066	6,147	7,210

Notes: [1/]The superscript *, **, and *** denote p < 0.1, 0.05, and 0.01 respectively.

Table 5 Effect of the Minimum Wage on Employment

Explanatory Variables	Dependent variable: Employment			
	2002-2011	2012-2013		
		Thai Workers	Foreign Workers	Overall
Minimum Wage	-0.0924**	-0.00561	-0.0753	-0.00759
	(0.0403)	(0.0145)	(0.0585)	(0.0138)
Female	0.0458	0.0414	0.169	0.0411
	(0.0457)	(0.128)	(0.356)	(0.121)
Years of Schooling	0.0334***	0.0165	-0.0163	0.0125
	(0.00837)	(0.0162)	(0.0606)	(0.0156)
Potential Working Experience	0.0379***	0.0648***	0.0837**	0.0630***
	(0.00622)	(0.0123)	(0.0423)	(0.0117)
Potential Working Exp.^2	-0.000502***	-0.000915***	-0.00124	-0.000896***
	(9.43e-05)	(0.000184)	(0.000843)	(0.000174)
Foreigner				1.113***
				(0.203)
Observations	142,073	18,822	961	21,744
Number of Panel ID	35,739	7,114	341	8,166

Notes: [1/]The superscript *, **, and *** denote p < 0.1, 0.05, and 0.01 respectively.

Table 6 The minimum wage elasticity of employment

Elasticity	2002-2011	2012-2013		
		Thai Workers	Foreign Workers	Overall
Minimum wage	-0.0450**	-0.0029	-0.0474	-0.0038
	(.01994)	(.0076028)	(0.0427616)	(.0069537)

Table 7 Effect of the Minimum Wage on Dis-employment

Explanatory Variable: Percent change in the minimum wage x Dummy variables for	Dependent variable: Dis-employment			
	2002-2011	2012-2013		
		Thai Workers	Foreign Workers	Overall
W/MW ≤ 0.9	0.0928	-0.109	3.580*	-0.0788
	(0.0686)	(0.0817)	(2.085)	(0.0762)
0.9 < W/MW ≤ 1.1	0.144**	-0.120	3.636*	-0.0914
	(0.0645)	(0.0752)	(2.096)	(0.0708)
1.1 < W/MW ≤ 1.2	0.0725	-0.121*	3.534*	-0.0940
	(0.0722)	(0.0710)	(2.079)	(0.0681)
1.2 < W/MW ≤ 1.3	0.124**	-0.128*	3.612*	-0.0990
	(0.0624)	(0.0728)	(2.089)	(0.0697)
1.3 < W/MW ≤ 1.5	0.0513	-0.114	3.454*	-0.0953
	(0.0488)	(0.0702)	(2.032)	(0.0672)
1.5 < W/MW ≤ 2	0.100	-0.0982	3.558*	-0.0709
	(0.0635)	(0.0601)	(2.087)	(0.0559)
2 < W/MW ≤ 3	0.138**	-0.112	3.529*	-0.0959
	(0.0574)	(0.0712)	(2.055)	(0.0686)
3 < W/MW ≤ 4	0.0997	-0.125	3.568*	-0.0975
	(0.0773)	(0.0785)	(2.083)	(0.0743)
4 < W/MW ≤ 5	0.195**	-0.118	3.534*	-0.0910
	(0.0959)	(0.0724)	(2.075)	(0.0710)
5 < W/MW ≤ 6	0.175**	-0.121	3.539*	-0.0940
	(0.0733)	(0.0776)	(2.084)	(0.0742)
6 < W/MW ≤ 8	0.274**	-0.125	3.581*	-0.0962
	(0.110)	(0.0768)	(2.094)	(0.0736)
Number of observations	152,187	10,311	3,245	13,556
Adjusted R-Square	0.011	0.058	0.006	0.040

Notes: [1/]The superscript *, **, and *** denote p < 0.1, 0.05, and 0.01 respectively.

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